

Anh Bertz

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EDUCATION

Georgia Institute of Technology

Master of Science in Computer Science. GPA: 4.0/4.0

Atlanta, GA

Jan. 2025 – May 2027

University of Wisconsin-Madison

Bachelor of Science in Computer Science. GPA: 3.64/4.0

Madison, WI

Sep. 2021 – Dec. 2023

EXPERIENCE

Software Developer

Phin Solutions, LLC

Sep. 2025 – Present

Ramsey, MN

- Designed or enhanced **20+ SQL stored procedures** and **queries** to speedup application functionalities by simplifying complex queries which improved reliability and efficiency of data access across production environments
- Converted **20+** types of legacy FoxPro to **C#** for core parsing logic which increased reliability, maintainability, and reduced execution time by 2x
- Built a multi-threaded **C# WinForms application** to increase process and export speed which ensured timely delivery of data allowing the business to acquire larger clients

Full Stack Developer

She Inspires Me

Apr. 2024 - Sep. 2025

Irving, TX

- Designed and implemented an admin panel allowing non-technical team members to contribute to the website content management without needing to dive into **Next.js** and **SCSS** source code, reducing content update time by an average of 4 days
- Developed and integrated **10+ C# APIs** used for updating website's content with data provided through a **MySQL** database, allowing for dynamic content loading and storage
- Implemented simple and scalable **SQL** database schemas to efficiently store user data, donations, and content management system data, emphasizing maintainability of the system

PROJECTS

Limit Order Book Streaming Service | *C#, WinForms, gRPC*

- Developed a **gRPC** client-server **C#** application with server streaming using a broadcast daemon for real-time order book updates and client unary calls for modifying order book state
- Implemented coarse-grained locking to manage concurrency for multiple clients and a **ConcurrentQueue** for receiving orders for order matching using a **First In, First Out** matching algorithm
- Structured limit order book levels using linked lists and Dictionaries for constant time modifications and removals of orders, and stored each level in a sorted set for efficient $O(\log_2(N))$ order insertions/matching

Enigma Machine Simulator | *Java, SQL, AWS RDS, JDBC, HTML, CSS, JavaScript, Springboot, Docker, JUnit*

- Created interactive UI features in **JavaScript** for dynamic display updates triggered by rotor and plugboard changes and custom keyboard handling to capture key presses for encryption and live ciphertext display
- Architected the core Enigma encryption algorithms and components in **Java**, including developing **JUnit** integration tests to validate correct encryption/decryption behavior
- Setup **AWS RDS MySQL** database to store, retrieve, and manipulate rotor configurations, messages, and user data, storing **1000+ user** records with **50+** unique rotor configurations

Virtual Reality Battle Chess AI | *C#, Unity*

- Designed and implemented a Chess AI engine using **C#** and machine learning algorithms, including **Monte Carlo Tree Search** and **Mini-max with alpha-beta pruning**
- Collaborated with 3 teammates to optimize default **VR** configurations for enhanced gameplay, including audio settings, movement tracking, and comfort adjustments

TECHNICAL SKILLS

Languages: C#, SQL, Java, Python, C/C++, JavaScript, HTML, CSS, R, Tableau, Power BI

Frameworks: Next.js, React, Node.js, JUnit, AWS RDS, Docker, Springboot, gRPC

Other: Git, JDBC, pandas, NumPy, Matplotlib, Excel, Word

Certificates: Google Data Analytics Specialization